



National  
Qualifications  
2018

**X747/76/12**

**Mathematics  
Paper 2**

THURSDAY, 3 MAY

10:30 AM – 12:00 NOON

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**Total marks — 70**

Attempt ALL questions.

**You may use a calculator.**

Full credit will be given only to solutions which contain appropriate working.

State the units for your answer where appropriate.

Answers obtained by readings from scale drawings will not receive any credit.

Write your answers clearly in the spaces provided in the answer booklet. The size of the space provided for an answer should not be taken as an indication of how much to write. It is not necessary to use all the space.

Additional space for answers is provided at the end of the answer booklet. If you use this space **you must clearly identify the question number** you are attempting.

Use **blue** or **black** ink.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.



## FORMULAE LIST

### Circle:

The equation  $x^2 + y^2 + 2gx + 2fy + c = 0$  represents a circle centre  $(-g, -f)$  and radius  $\sqrt{g^2 + f^2 - c}$ .

The equation  $(x - a)^2 + (y - b)^2 = r^2$  represents a circle centre  $(a, b)$  and radius  $r$ .

### Scalar Product:

$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$ , where  $\theta$  is the angle between  $\mathbf{a}$  and  $\mathbf{b}$

or  $\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$  where  $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$  and  $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ .

### Trigonometric formulae:

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$= 2 \cos^2 A - 1$$

$$= 1 - 2 \sin^2 A$$

### Table of standard derivatives:

| $f(x)$    | $f'(x)$      |
|-----------|--------------|
| $\sin ax$ | $a \cos ax$  |
| $\cos ax$ | $-a \sin ax$ |

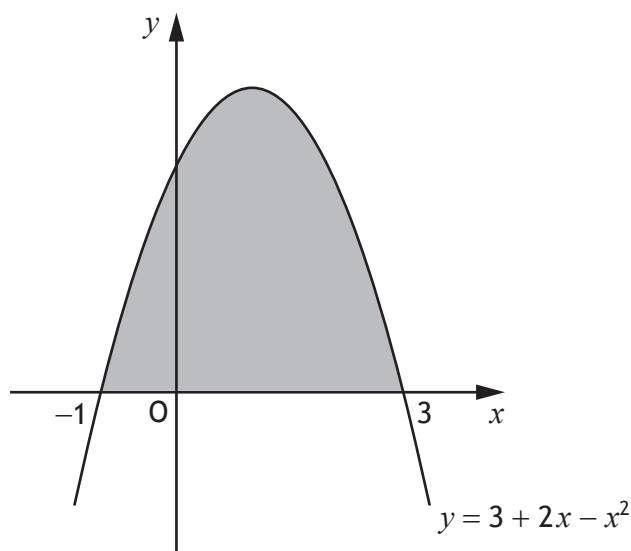
### Table of standard integrals:

| $f(x)$    | $\int f(x) dx$             |
|-----------|----------------------------|
| $\sin ax$ | $-\frac{1}{a} \cos ax + c$ |
| $\cos ax$ | $\frac{1}{a} \sin ax + c$  |

Attempt ALL questions

Total marks — 70

1. The diagram shows the curve with equation  $y = 3 + 2x - x^2$ .



Calculate the shaded area.

4

2. Vectors  $\mathbf{u}$  and  $\mathbf{v}$  are defined by  $\mathbf{u} = \begin{pmatrix} -1 \\ 4 \\ -3 \end{pmatrix}$  and  $\mathbf{v} = \begin{pmatrix} -7 \\ 8 \\ 5 \end{pmatrix}$ .

(a) Find  $\mathbf{u} \cdot \mathbf{v}$ .

1

(b) Calculate the acute angle between  $\mathbf{u}$  and  $\mathbf{v}$ .

4

3. A function,  $f$ , is defined on the set of real numbers by  $f(x) = x^3 - 7x - 6$ .

Determine whether  $f$  is increasing or decreasing when  $x = 2$ .

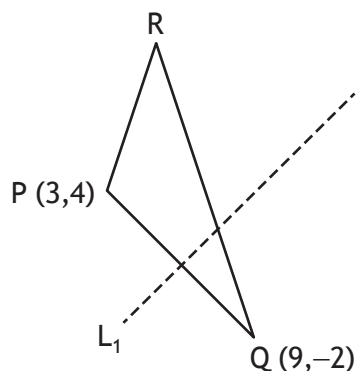
3

4. Express  $-3x^2 - 6x + 7$  in the form  $a(x+b)^2 + c$ .

3

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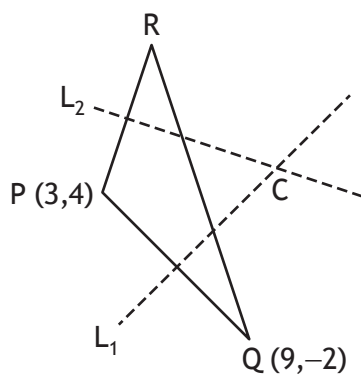
5. PQR is a triangle with  $P(3,4)$  and  $Q(9,-2)$ .



- (a) Find the equation of  $L_1$ , the perpendicular bisector of PQ.

3

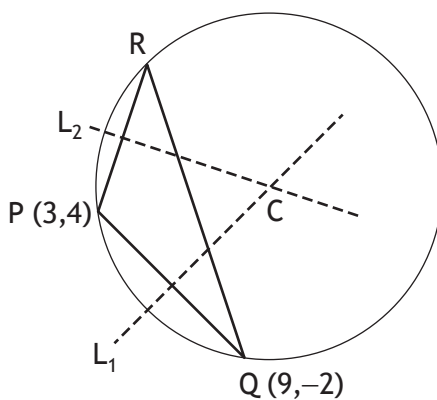
The equation of  $L_2$ , the perpendicular bisector of PR is  $3y + x = 25$ .



- (b) Calculate the coordinates of C, the point of intersection of  $L_1$  and  $L_2$ .

2

C is the centre of the circle which passes through the vertices of triangle PQR.



- (c) Determine the equation of this circle.

2

6. Functions,  $f$  and  $g$ , are given by  $f(x) = 3 + \cos x$  and  $g(x) = 2x$ ,  $x \in \mathbb{R}$ .

(a) Find expressions for

(i)  $f(g(x))$  and

2

(ii)  $g(f(x))$ .

1

(b) Determine the value(s) of  $x$  for which  $f(g(x)) = g(f(x))$  where  $0 \leq x < 2\pi$ .

6

7. (a) (i) Show that  $(x-2)$  is a factor of  $2x^3 - 3x^2 - 3x + 2$ .

2

(ii) Hence, factorise  $2x^3 - 3x^2 - 3x + 2$  fully.

2

The fifth term,  $u_5$ , of a sequence is  $u_5 = 2a - 3$ .

The terms of the sequence satisfy the recurrence relation  $u_{n+1} = au_n - 1$ .

(b) Show that  $u_7 = 2a^3 - 3a^2 - a - 1$ .

1

For this sequence, it is known that

- $u_7 = u_5$
- a limit exists.

(c) (i) Determine the value of  $a$ .

3

(ii) Calculate the limit.

1

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8. (a) Express  $2\cos x^\circ - \sin x^\circ$  in the form  $k\cos(x-a)^\circ$ ,  $k > 0$ ,  $0 < a < 360$ . 4

(b) Hence, or otherwise, find

(i) the minimum value of  $6\cos x^\circ - 3\sin x^\circ$  and 1

(ii) the value of  $x$  for which it occurs where  $0 \leq x < 360$ . 2

9. A sector with a particular fixed area has radius  $x$  cm.

The perimeter,  $P$  cm, of the sector is given by

$$P = 2x + \frac{128}{x}.$$

Find the minimum value of  $P$ .

6

10. The equation  $x^2 + (m-3)x + m = 0$  has two real and distinct roots.

Determine the range of values for  $m$ .

4

11. A supermarket has been investigating how long customers have to wait at the checkout.

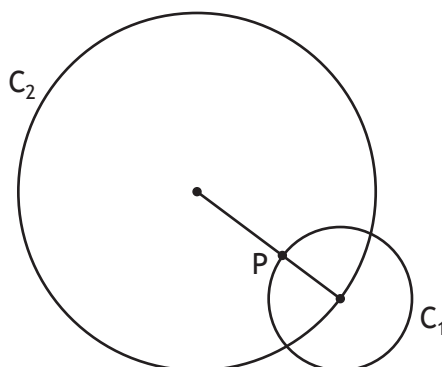
During any half hour period, the percentage,  $P\%$ , of customers who wait for less than  $t$  minutes, can be modelled by

$$P = 100(1 - e^{-kt}), \text{ where } k \text{ is a constant.}$$

(a) If 50% of customers wait for less than 3 minutes, determine the value of  $k$ . 4

(b) Calculate the percentage of customers who wait for 5 minutes or longer. 2

12. Circle  $C_1$  has equation  $(x-13)^2 + (y+4)^2 = 100$ .  
 Circle  $C_2$  has equation  $x^2 + y^2 + 14x - 22y + c = 0$ .



- (a) (i) Write down the coordinates of the centre of  $C_1$ . 1  
 (ii) The centre of  $C_1$  lies on the circumference of  $C_2$ .  
 Show that  $c = -455$ . 1

The line joining the centres of the circles intersects  $C_1$  at P.

- (b) (i) Determine the ratio in which P divides the line joining the centres of the circles. 2  
 (ii) Hence, or otherwise, determine the coordinates of P. 2

P is the centre of a third circle,  $C_3$ .

$C_2$  touches  $C_3$  internally.

- (c) Determine the equation of  $C_3$ . 1

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